Tezpur University
End semester examination (Spring 2024)
CO218: Data Communication

Full marks=60 *Time=3 hours*

(The figures in the right-hand margin indicate full marks for the questions)

1. Answer the following? (any **five**) **[2x5=10]**
   a. State the bidding and conflict avoidance rule in binary count down protocol.
   b. "Stop and wait becomes inadequate, with the use of multiple frames for a single message". Justify.
   c. What are the network layer service design goals?
   d. How does MACAW overcome the issue faced by the MACA protocol?
   e. What is the count-to infinity problem faced by distance vector routing algorithm?
   f. Write the difference between the two types of bridges.
   g. How is IEEE 802.5 different from IEEE 802.4?
   h. Write the difference between router and gateway.

2. What do you understand about VLAN trunking? Name the types of VLAN? **[1+2=3]**

3. What are the advantages offered by spread spectrum technique over fixed frequency transmission? Name the classes of signature sequences used by CDMA. **[2+1=3]**

4. How is "listen before talk" policy is implemented by CSMA? In what situation collision is still possible in CSMA/CD? **[2+1=3]**

5. Discuss about the channel acquisition strategies adopted in cable network? Name the classes of channel access mechanisms implemented by IEEE 802.11 MAC protocol? **[3+1=4]**

6. Discuss briefly how are Go Back N and Selective Repeat achieved in HDLC? How does Go Back N handle the situation when an ACK to the sender is lost in between? **[2+2+1=5]**

7. Calculate the polynomial code checksummed frame for a frame 1101011011 using the generator polynomial $x^4 + x^2 + 1$. **[3]**

8. How are flow and error control done in TDM link? Why is the statistical TDM important over TDM link? **[3]**

9. A time division multiplexer multiplexes four 100 bps channels. If the MUX multiplexes 1 byte per channel, compute the frame size. rate, duration and the bit rate of the link. **[4]**

10. A system is using NRZ-I to transfer 1-Mbps data. What are the average signal rate and minimum bandwidth? **[4]**

11. Discuss about different types of operational modes implemented by IEEE 802.2 standard. **[4]**

Or

12. List the drawbacks of distance vector routing algorithm. How does link state routing algorithm solve the issues? **[4]**

13. Consider four digital data channels each transmitting at 1 Mbps speed. Design an appropriate configuration for FDM, using a satellite channel of bandwidth 1 MHz. **[4]**

14. Consider a carrier signal needs to carry user's data in which one data element is encoded as two signal elements with the bit rate 10 kbps. Calculate the average value of the baud rate if case factor remains in between 0 and 2. **[4]**

15. Answer the following questions appropriately. (any **three**) **[2x3=6]**
    a) Draw the waveform to encode the data string "0100010101100000000011" using HDB3 encoding scheme
    b) What is the advantage of sliding-window flow control compared to stop-and-wait flow control?
    c) How does slotting improve performance of ALOHA?
    d) Name the control signaling virtual channel connection (VCC) implemented by ATM networks
    e) Write two advantages of OFDM technique.

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